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COS720
Research and Design Report
Computer and Information Security I
Insider Threat Detection
InsiderEyeShield



19 May 2026

Generative AI declaration form for assignment submission

Q1. Student Information

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Course Code	COS720
Assignment Title	Insider Threat Detection

Q2. AI Declaration [tick one box]

1. I used Generative Artificial Intelligence in the current assignment	X
2. I did not use any Gen AI for the current assignment	

Ensure you are permitted to use generative AI by reading the instructions for your assignment or assessment, or by consulting your study guide. Failure to follow the instructions regarding using generative AI for your assignment constitutes academic dishonesty. Examples of generative AI include ChatGPT, Google's Gemini, Microsoft's Copilot, Claude, and Meta's Llama 3 AI integrated into WhatsApp and Facebook. If you selected option 1 in Q1, please continue to complete the rest of the form:

Q3. Prompts used

Please paste all the prompts you created for the assignment and indicate which aspect(s) of the assignment the specific prompt was used for.

AI Tool	Prompt	Aspect of Assignment
Claude	"Can You generate a project timeline for me?"	Time Management
ChatGPT	"Can you generate a logo for InsiderEyeShield?"	Logo and colour coding
Claude	"What application can I use to run my prototype that can handle data and analytics well? "	Application of a prototype
Claude	"Benefits of using Recall over F1 Score in IDS"	Clear understanding

Q4. Type of usage

Briefly describe the features for which you used the Gen AI. You may say "Not Applicable" where needed:

Feature used	Description
Brainstorming and idea generation	X UIX Design
Language editing suggestions	
Feedback and revision suggestions	
Explaining complex concepts	X Mentioned Above
Writing coach	X Grammarly
Other (please specify):	

Q5. Ethical use

Write a brief paragraph explaining how you ensured the usage of Gen AI was aligned with the ethical and responsibility requirements of the University of Pretoria ([link](#)). Consider examples such as repurposing and reintegrating ideas generated by Gen AI with your own thoughts, integrating Gen AI ideas with other literature, critically evaluating Gen AI outputs, maintaining transparency about Gen AI usage, enhancing your learning and ensuring comprehension despite using Gen AI, and personal

development through using Gen AI as an assistant. If you directly used text or data generated by Gen AI, ensure it was cited appropriately.

Ethical use statement:

I ensured that my use of Generative AI aligned with the ethical and responsible use guidelines of the University of Pretoria by using AI as a support tool to enhance my understanding and productivity rather than replace my own work. These outputs were critically evaluated and adapted to fit the requirements of the project and were integrated with my own ideas, implementation, and supporting literature. Additionally, I used Grammarly to proofread and improve the clarity and professionalism of my writing. Throughout the project, I maintained transparency regarding the use of AI tools and ensured that all final decisions, interpretations, and submitted work reflected my own understanding and learning, in accordance with academic integrity requirements.

Q6. Why did you use Gen AI?

I used AI to help interpret and resolve coding errors, better understand evaluation metrics and model performance, and generate UI/UX imagery for interface design concepts, which expanded my knowledge on new systems as well as helped spark creativity in developing the application. And building UIX that was not off-putting nor taking away from the workings of the application

Signature:

Student Signature	WO Molokwe
Date	19/05/2026

Note: The Department of Education Innovation would like to thank Prof. Celeste Combrinck and Ms. Nelé Loubser for their innovative ideas, which have been instrumental in creating our new form.

Contents

1	Introduction	3
2	Research Overview	3
2.1	What is Insider Threat ?	3
2.2	Artificial Intelligence and Machine Learning in Behavioural Threat Detection	3
2.3	Rule-Based VS AI-Driven Detection	3
2.4	Summary of Shoderu et al. Paper	3
3	Requirements Specification	4
3.1	Functional Requirements	4
3.2	Non-Functional Requirements	4
4	Feature Selection and Dataset Justification	5
5	Model Selection: Random Forest + Logistic Regression Ensemble	6
6	UML Designs	6
6.1	Use Case Diagram	6
6.2	System Architecture	7
6.3	Class Diagram	7
6.4	Use Case Diagram	7
7	Testing Summary	7
7.1	Test Configurations	7
7.2	Confusion Matrix	8
7.3	Model Specific Testing	8
7.4	Limitations and Ethical Considerations	9
7.5	Limitations	9
7.6	Ethical Considerations	9

1 Introduction

"Insider Threat is a security risk that occurs when a current or former personnel, contractor, or other business partner who has or had authorised access to an organisation's network infrastructure, systems, or data and who intentionally (or negligently) exceeds or misuses the access to harm an organisation " [1].

2 Research Overview

2.1 What is Insider Threat ?

Insider Threat is one of the most demanding threats in cybercrimes. [2]

The challenging part of this threat is the human component, as it can be difficult to understand a person's intentions. Unlike external attackers, who would need to brute-force or scheme their way to obtain access, the attackers already have valid credentials, knowledge of the organisation's processes and procedures, and privileges to certain systems.

These threats cause a substantial loss in an organisation and pose a major security risk. As data and information increase, their value increases as well, making threats such as insider threats increase. [3]

Therefore, Cyber Security professionals and organisations have discovered a way to prevent or mitigate these attacks [4], whether intentional or unintentional, by utilising security controls and policies.

[4]However, even with these controls and policies, a survey found that 68% of the security professionals responded that they were moderately to extremely susceptible to insider threats.

2.2 Artificial Intelligence and Machine Learning in Behavioural Threat Detection

The use of Artificial Intelligence (AI) and Machine Learning (ML) techniques provides essential components to insider threat detection, instead of relying on fixed rules that outline known bad traits.

Algorithms such as Random Forest [5], PCA which was used with Deep Feature Synthesis for behavioural feature extraction [6], Logistic Regression , Convolutional Neural Networks(CNNs) [7], Support Vector Machine (SVM) with LSTM for feature extraction [8] [9], etc, are trained on labeled historical data to "learn" and plot the boundaries between normal and malicious behaviour.

Random Forest [5] excels in recording complex, non-linear inter-dependence of behavioural features, while other algorithms, such as Logistic Regression, capture linear decision boundaries that are resistant to overfitting on smaller data and are easy to interpret.

2.3 Rule-Based VS AI-Driven Detection

Rule-based detection models depend on pre-defined rules or policies to single out suspicious activities in an organisation. They often utilise regular expressions to search for specific patterns in log files, [9]. They need to be manually updated and may get too complex and lose scalability with the volume of data over time, and threats may be missed if they do not match pre-defined rules.

AI-driven detections adapt easily over time and evolve with behaviour automatically, and learn from trained data. They work better with an increase in data, as well as support explainability by the use of feature importance scores, breaking down risk factors and confidence probabilities, which allow security analysts to understand the reasoning behind flagging an individual.

2.4 Summary of Shoderu et al. Paper

[10]Even with these developments, studies, and models, there is an absence of a well-structured Digital Forensic Readiness (DFR) framework with Software-as-a-Service(SaaS) environments. Furthermore, the strategies of mitigation lean more towards reactive than proactive. Meaning that the SOC department in an organisation begins to collect evidence after an incident has occurred, which can be problematic in a SaaS environment due to evidence being fragmented over provider-controlled infrastructure.

Shoderu et al. proposed the use of a five-component framework called the Digital Forensic Readiness to Bust

Insider Threats (DFR-BUST), which aligns with the ISO/IEC 27043 standards. The components include SaaS Evidence Preparation (SEP), Detection Engine Process (DEP), Evidence Archival and Preservation (EAP), Analyst and Forensic Review Process (AFR), and Audit and Documentation Process (ADP).

The study showcases how proactive evidence preparation, combined with AI-based behavioural analytics, improves the investigative process and enables early recognition of insider threat activity, while preserving forensic soundness.

However, the existing detection models and research overwhelmingly centre around controlled laboratory environments. This study focuses on providing a prototype of an operational Security Operations Centre Interface through which a security analyst can investigate, triage, and act on the outputs provided by AI-driven models trained on a realistic labeled dataset.

3 Requirements Specification

3.1 Functional Requirements

FR.1: User Assessment

- FR.1.1. System shall classify a user's behaviour record as either malicious
- FR.1.2. System shall output a confidence probability for each classification
- FR.1.3. System shall assign a risk level (Low/ Medium/ High/ Critical)

FR.2: SOC Dashboard :

- FR.2.1 System shall provide a SOC Dashboard to display threat metrics
- FR.2.2 System shall allow user view flagged employees
- FR.2.3 System shall allow a user to view risk distribution per department

FR.3: Analytics and Live Monitoring

- FR.3.1 System shall allow user to view analytics at department level
- FR.3.2 System shall allow user to view heatmaps
- FR.3.3 System must provide a simulated real-time event feed
- FR.3.4 System must provide rolling risk trend Sparkline

FR.4: Reporting:

- FR.4.1 System shall provide a report with executive KPIs
- FR.4.2 System shall provide model performance metrics, confusion matrices.
- FR.4.3 System shall allow a user to download a CSV report.

FR.5: Investigative

- FR.5.1 System shall allow users to manually insert employee and profiles
- FR.5.2 System shall allow user to view stored flagged profiles
- FR.5.3 System shall display AI explainability outputs, feature importance rankings, risk scores, and recommendations.

3.2 Non-Functional Requirements

QR.1: Performance :

- QR.1.1 System shall achieve a minimum Recall of 75% on the test set to minimize missed threats.
- QR.1.2 System shall load and render all pages within 5 seconds on a standard development machine.
- QR.1.3 System shall load all analytics clearly, with correct colour codes
- QR.1.4 System shall load all model artifacts within 5 seconds on a standard development machine

QR.2: Usability :

- QR.2.1 System shall provide risk explanations that are understandable to a non-technical security manager without specialist ML knowledge.

QR.2.2 System shall provide clear visual differentiation between risk levels using consistent color codes.

QR.2.3 System shall be easy to navigate using navigation bars

QR.3: Reliability :

QR.3.1 System shall load model artifacts (trained model, scaler, encoders) which are serialized and reloaded at startup without retraining.

QR.3.2 System must consistently utilize model without breaking.

4 Feature Selection and Dataset Justification

The dataset used in this study uses the Kaggle dataset *Insider Threat Dataset for Behavioral Features & Threat Classification Dataset for Security Assessment* [11].

The dataset contains employee behavioural patterns and indicators for 118,614 records from a corporate environment. It contains 21 features of which 17 are numerical and 4 categorical. There are no missing, inf or NaN values in the dataset, as well as no duplicates. It covers 3 different campuses, 11 departments, 48 positions and 32 countries.

An Exploratory Data Analysis is conducted on the dataset, to explore the dataset to view relations within the datasets, with this overall output:

Dataset : 111,027 records $\times$ 22 features

Target : is_malicious

Benign : 108,056 (97.3%)

Malicious : 2,971 (2.7%)

Imbalance : 17.6:1

Top Pearson Correlates with target:

- total_files_burned $r = 0.423$
- burned_from_other $r = 0.297$
- num_printed_pages_off_hours $r = 0.197$
- total_printed_pages $r = 0.156$
- num_unique_campus $r = 0.123$
- hostility_country_level $r = 0.109$
- entry_during_weekend $r = 0.105$
- num_entries $r = 0.102$
- employee_classification $r = 0.050$
- employee_seniority_years $r = 0.045$

Top RF Feature Importances:

- total_files_burned 0.2238
- total_printed_pages 0.1298
- employee_seniority_years 0.1269
- employee_position 0.1065
- employee_origin_country 0.0981

The dataset is then split with 80/20 for training and testing. During the split, the data is stratified to preserve the malicious dataset. The ratio of the imbalance is 17.6: 1, to balance this SMOTE is applied only to the training set of the dataset, where synthetic data is generated in order to balance the target feature.

The categorical features are then encoded using LabelEncoder, this is to reduce the number of encoders needed for One-Hot Encoding, as well as for categories such as Positions in an organization, order is important.

All features are preserved besides the '**late_exit_flag**' feature during the EDA, it is noted that there is no variance. and due to the rarity of the occurrence of the threat, the rest of the features play a part in training, to ensure the model does not miss threats. Some features are behavioural, temporal, show access behaviour, privilege, provide background.

5 Model Selection: Random Forest + Logistic Regression Ensemble

The model selection process is evaluated among Logistic Regression (LR), Random Forest (RF), Decision Tree (DT), combination of LR + RF, combination of LR + DT and a combination of all three.

The models are trained for 10 runs, over different seeds, the training dataset is shuffled on every run to avoid the models 'cramming' the dataset. The model is selected based on the average recall of the model rather than the F1 Score, this is due to the cost in insider threat detection that favours minimizing false negative (missed real threats) over false positives (false alarms that analysts can clear).

6 UML Designs

6.1 Use Case Diagram



Figure 1: Use Case Diagram

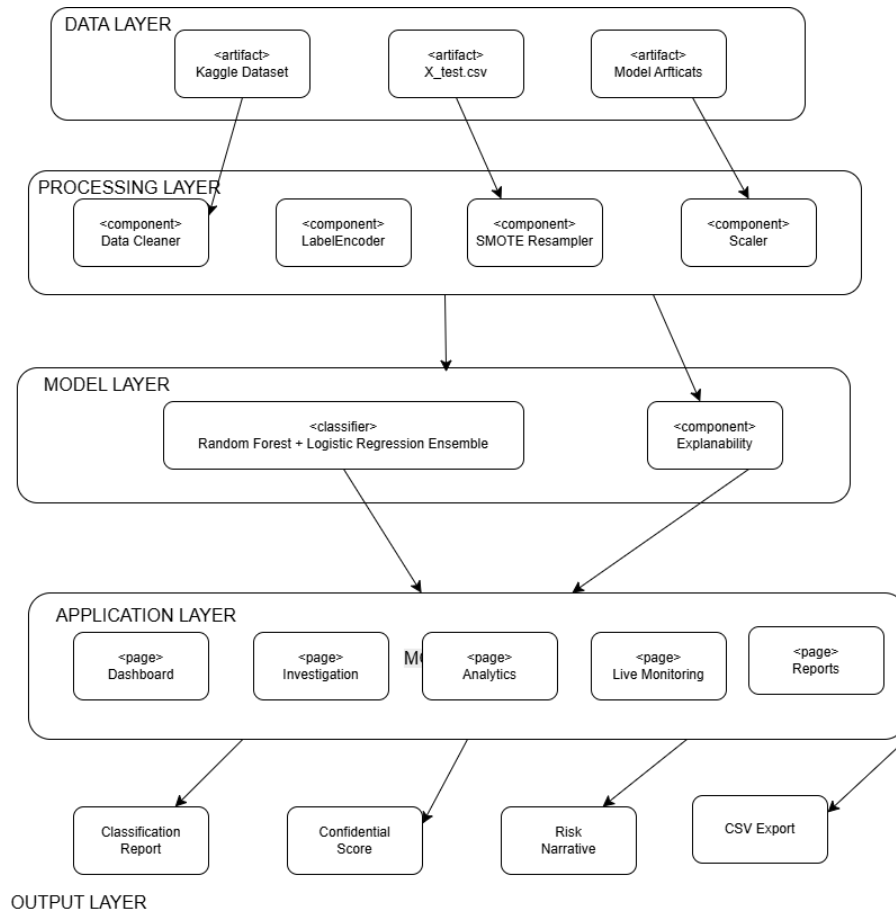


Figure 2: System Architecture Diagram

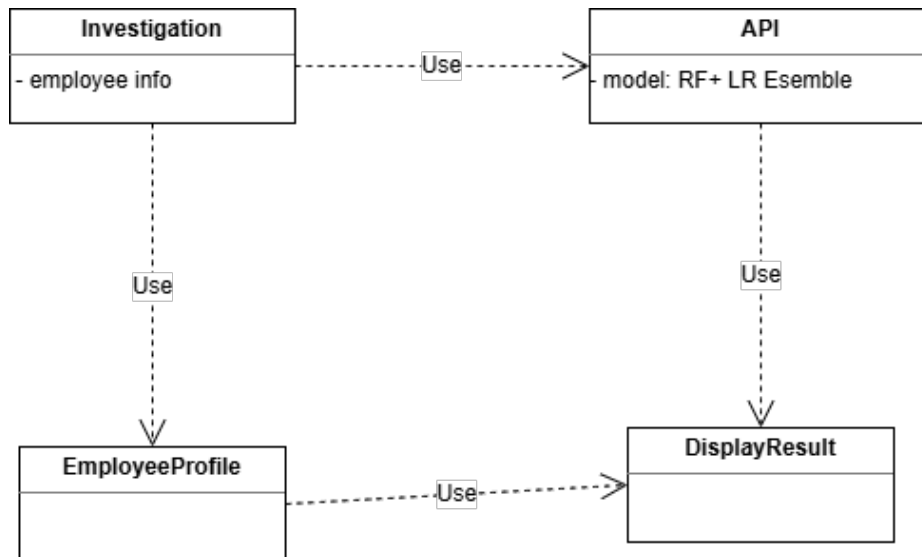


Figure 3: Class Diagram for the investigation

6.2 System Architecture

6.3 Class Diagram

6.4 Use Case Diagram

7 Testing Summary

7.1 Test Configurations

The model is tested on 20% of the stratified test set.

7.2 Confusion Matrix

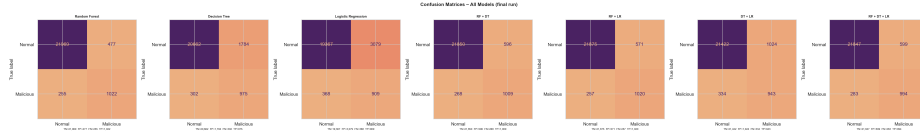


Figure 4: Confusion Matrix Results

7.3 Model Specific Testing

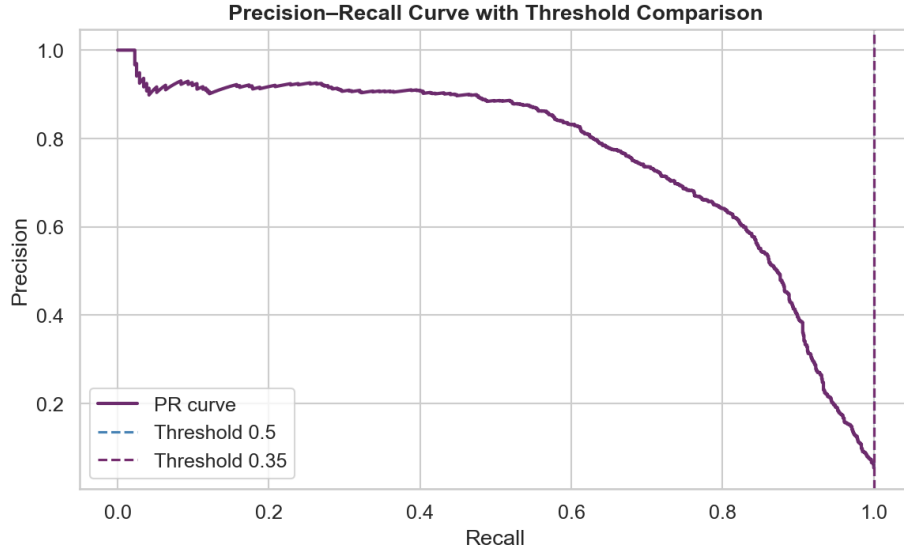


Figure 5: Precision-Recall Curve for RF+LR Ensemble

This curve displays the that precision and recall drops when the decision threshold of the attack is set to 0.35, but optimizes at threshold of 0.5.

Table 1: Classification Report: RF + LR Ensemble (Best by Recall = 0.8002)

Class	Precision	Recall	F1-Score	Support
Normal (0)	0.99	0.97	0.98	22,446
Malicious (1)	0.64	0.80	0.71	1,277
Accuracy		0.97		23,723
Macro avg	0.81	0.89	0.85	23,723
Weighted avg	0.97	0.97	0.97	23,723

Confusion Matrix	Count
True Negatives (correct Normal)	21,875
False Positives (Normal - Malicious)	571
False Negatives (Malicious - Normal)	257
True Positives (correct Malicious)	1,020

The classification table documents the true positives, false positives and false negatives, the number of false positives is a bit high, and in this regard the model performs a bit poorly, however the ability of the model to detect more true negatives and true positives is significant. To improve the model, more data is needed to train the model, if applicable transfer learning can also be utilized to increases how well the model learns.

7.4 Limitations and Ethical Considerations

7.5 Limitations

- The model is trained over synthetic data, which may not generalize in a real-world environment.
- The web platform only simulates live monitoring rather than consume live event stream.
- With a recall of 80% that means 20% of real threats are missed, therefore the system can be used as decision supporting tool.

7.6 Ethical Considerations

- There are privacy concerns that are being raised by continuous monitoring, and deployment need to follow organization policy and the employee needs to be aware of this monitoring.
- Features such as *has_foreign_citizenship*, *hostility_county_level* introduce bias risk and should be included with caution before deployment
- Data retention needs to follow the POPIA and GDPR act when handling personal data
- InsiderEyeShield is optimized for Recall which means organizations must redesign and review workflows that protect employee data from false alarms.

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